

Lecture 7 · Uncertainty I: Expected Utility and Risk Aversion

MWG Ch. 6.A–6.C · utility finally goes cardinal

Advanced Microeconomics 1

Thursday 22 October

Why uncertainty, and why now

- Everything so far assumed choices have *known* consequences. Almost nothing interesting does: portfolios, insurance, careers, crops.
- The move: alternatives become **lotteries** — probability distributions over outcomes — and we ask what rational preference over lotteries looks like.
- The payoff is structural: one new axiom (independence) buys a representation that is **linear in probabilities**, and with it something ordinal utility never had — **cardinal content**. Curvature of u will finally *mean* something: risk aversion.
- And a warning up front: this is the one block of the course where the axioms fail *in your own data* — we get to that at the end.

Lotteries and the independence axiom

Lotteries

Outcomes $C = \{c_1, \dots, c_N\}$ (finite for today).

Definition · simple and compound lotteries (MWG 6.B.1–6.B.2)

A **simple lottery** is $L = (p_1, \dots, p_N)$, $p_n \geq 0$, $\sum_n p_n = 1$. A **compound lottery** $(L_1, \dots, L_K; \alpha_1, \dots, \alpha_K)$ yields simple lottery L_k with probability α_k .

- Maintained hypothesis (*consequentialism*): only the **reduced** simple lottery matters — $\sum_k \alpha_k L_k$. How uncertainty resolves, in how many stages, is irrelevant. Already substantive: rules out anxiety, suspense, preference for one-shot vs. gradual resolution.
- Geometry: the set of simple lotteries is the simplex Δ^{N-1} ; with $N = 3$, a triangle — the canvas for everything today.

The axioms

Preferences $\succsim$ on the space of simple lotteries $\mathcal{L}$, assumed rational, plus:

Definition · continuity (MWG 6.B.3)

For all L, L', L'' : the sets $\{\alpha \in [0, 1] : \alpha L + (1 - \alpha)L' \succsim L''\}$ and $\{\alpha : L'' \succsim \alpha L + (1 - \alpha)L'\}$ are closed. (No lexicographic-style jumps; small probability changes don't reverse strict rankings.)

Definition · independence (MWG 6.B.4)

For all L, L', L'' and $\alpha \in (0, 1)$:

$$L \succsim L' \iff \alpha L + (1 - \alpha)L'' \succsim \alpha L' + (1 - \alpha)L''.$$

- Mixing both sides with a common third lottery cannot reverse a ranking: the L'' -branch happens or it doesn't, and if it doesn't you face the original comparison.

No analogue exists in consumer theory — there, mixing commodity bundles creates new bundles, not new bundles. In consumer theory, mixing bundles creates new bundles, not new bundles. In consumer theory, mixing bundles creates new bundles, not new bundles.

The Expected Utility Theorem

Theorem · v.N–M representation (MWG 6.B.2–6.B.3)

$\succsim$ on $\mathcal{L}$ is rational, continuous, and satisfies independence **iff** there exist numbers $u_1, \dots, u_N$ such that

$$L \succsim L' \iff \sum_n p_n u_n \geq \sum_n p'_n u_n.$$

Moreover $\tilde{U}$ represents the same $\succsim$ in expected-utility form **iff** $\tilde{u}_n = \beta u_n + \gamma$, $\beta > 0$:
unique up to positive affine transformations.

- Linearity in probabilities is the whole content. In the simplex: indifference curves are **parallel straight lines**.
- The uniqueness clause is the cardinality: ratios of utility *differences* are meaningful. “The gain from c_1 over c_2 is twice that from c_2 over c_3 ” now says something — it predicts choices among lotteries.

Proof sketch: the calibration construction

Assume best and worst outcomes $\bar{L} \succ \underline{L}$ (else trivial). Three steps:

1. **Monotonicity in the mixing weight:** $\beta \bar{L} + (1 - \beta) \underline{L} \succ \gamma \bar{L} + (1 - \gamma) \underline{L}$ iff $\beta > \gamma$ — from independence.
2. **Calibrate:** by continuity, for each lottery L there is a unique $\alpha_L \in [0, 1]$ with

$$L \sim \alpha_L \bar{L} + (1 - \alpha_L) \underline{L}. \quad \text{Define } U(L) = \alpha_L.$$

3. **Linearity:** independence forces $\alpha_{\beta L + (1 - \beta)L'} = \beta \alpha_L + (1 - \beta) \alpha_{L'}$ — substitute each lottery by its calibrated equivalent inside the mixture. $\square$ (sketch)
 - Compare Lecture 2's Debreu construction: measure a bundle by *where its indifference curve crosses the diagonal*; measure a lottery by *its indifference mixture of best and worst*. Same trick, new clothes — and this one delivers cardinality because mixing is linear.

Allais: independence meets the lab

Choose in each pair (amounts in millions):

Pair 1:	L_1 : \$1 for sure	vs.	L'_1 : \$5 w.p. .10, \$1 w.p. .89, \$0 w.p. .01
Pair 2:	L_2 : \$1 w.p. .11, \$0 w.p. .89	vs.	L'_2 : \$5 w.p. .10, \$0 w.p. .90

- Modal pattern: $L_1 \succ L'_1$ and $L'_2 \succ L_2$. Subtract the common consequence (\$1 w.p. .89 vs. \$0 w.p. .89) and independence says the two pairs are *the same choice*. The modal pattern violates it.
- In the simplex: indifference lines that **fan out** instead of staying parallel. Responses: certainty effects, rank-dependent weighting, prospect theory (pointers Monday).
- MWG's own verdict is worth quoting in spirit: a theory this useful is not discarded for failing a boundary experiment — but you must know *where* the boundary is.

Money lotteries and risk aversion

Risk aversion = concavity

Outcomes now amounts of money; lottery = distribution F ; Bernoulli utility $u(x)$, continuous and increasing; $U(F) = \int u(x) dF(x)$.

Definition · risk aversion (MWG 6.C.1)

The agent is **risk averse** if for every F : $u(\int x dF) \geq \int u(x) dF$ — the expected value for sure beats the lottery. By Jensen's inequality, this holds **iff u is concave**.

Two equivalent measurements:

- **Certainty equivalent** $c(F, u)$: $u(c(F, u)) = \int u dF$; risk aversion $\iff c(F, u) \leq \int x dF$.
- **Risk premium** $\pi = \int x dF - c(F, u) \geq 0$: what the agent pays to shed the risk — the insurance industry's revenue, in one symbol.

Arrow–Pratt: how curved, in usable units

Definition · coefficients of risk aversion (MWG 6.C.2)

$$r_A(x) = -\frac{u''(x)}{u'(x)} \quad (\text{absolute}), \quad r_R(x) = -\frac{x u''(x)}{u'(x)} \quad (\text{relative}).$$

- Why this ratio: u'' alone is meaningless (affine transformations rescale it); dividing by u' makes r_A invariant — the *first* cardinal statistic of the course.
- Where it earns its units: for a small risk ε with mean 0 and variance σ^2 around x ,

$$\pi \approx \frac{1}{2} \sigma^2 r_A(x)$$

(second-order Taylor of both sides of the CE definition — a two-line derivation you must do cold).

- **CARA:** $u = -e^{-ax}$, $r_A \equiv a$ (wealth-invariant absolute). **CRRA:** $u = x^{1-\sigma}/(1-\sigma)$, $r_R \equiv \sigma$ (wealth-invariant relative — why growth theory scales, and why every macro model is CRRA).

Comparing people, comparing wealth levels

Proposition · Pratt (1964); MWG 6.C.2

The following are equivalent: (i) $r_A^2(x) \geq r_A^1(x)$ for all x ; (ii) $u_2 = \psi(u_1)$ for some increasing concave ψ ; (iii) $c(F, u_2) \leq c(F, u_1)$ for every F ; (iv) whenever agent 2 accepts a gamble, so does agent 1.

- One local statistic (r_A) controls a global behavioral ordering — the reason Arrow–Pratt is *the* language of applied risk work.
- Wealth effects: under **decreasing** absolute risk aversion (DARA — the empirically sane case), richer agents accept more absolute risk; the risky asset is a normal good in the classic portfolio problem (MWG 6.C.4–6.C.5, EC3).

How risk averse are people? Your data, and the field's

Two questions you answered in September

In the Session 1 quiz, before any of this theory:

Q1 50–50: lose €100 / gain €110. Accept?

Q2 50–50: lose €1,000 / gain €10,000,000. Accept?

- Your answers: [insert: n rejected Q1; of those, m accepted Q2].
- Hold that pattern on screen. Under expected utility, it is about to become a problem.

Rabin's calibration theorem

Theorem · Rabin (Econometrica 2000), flavor

Suppose an expected-utility maximizer (any increasing concave u) rejects the 50–50 lose \$100 / gain \$110 gamble *at every wealth level*. Then they reject a 50–50 lose \$1,000 / gain *any sum* gamble.

- Mechanism, in one breath: rejecting the small bet at w bounds $u'(w + 110) \leq \frac{100}{110} u'(w - 100)$; rejecting it *everywhere* compounds that decay geometrically, so marginal utility collapses over a few thousand dollars — and no finite prize can compensate a \$1,000 loss.
- So EU cannot explain **both** of: meaningful small-stakes risk aversion *and* sane large-stakes behavior. Concavity of u over wealth is a theory of *large* risks only.
- Your class pattern — reject small, accept large — is not a preference an EU maximizer can have. Candidate diagnoses: **loss aversion** and **narrow bracketing** (Rabin–Thaler 2001): the small gamble is evaluated in isolation, gains vs. losses, not merged into lifetime wealth.

The field agrees with the lab

Application · Sydnor (AEJ: Applied 2010) — over-insuring modest risks

Homeowners choosing deductibles routinely pay premia to reduce the deductible by a few hundred dollars against claim probabilities of a few percent — choices only astronomically large risk aversion could rationalize under EU on wealth. Small-stakes insurance is Rabin's theorem with a price tag.

Application · Chetty (AER 2006) — bounding σ from labor supply

Risk aversion and the curvature of consumption utility are the same parameter; observed labor-supply responses to wage changes bound it — implying *modest* curvature (low single digits), far below what small-stakes choices imply. The tension is the point.

- Working rule you will use for a career: **CRRA σ in the low single digits for large risks**; do not ask concave- u -over-wealth to explain lottery tickets or phone insurance.

Takeaways and readings

Takeaways

- Independence $\Rightarrow$ EU: linear in probabilities, unique up to positive affine transformations — utility is finally cardinal.
- Risk aversion $\iff$ concave u ; Arrow–Pratt makes curvature a usable statistic: $\pi \approx \frac{1}{2}\sigma^2 r_A$; Pratt's theorem makes “more risk averse” a global ordering.
- Calibration discipline: EU is a theory of large risks. Small-stakes risk aversion, including yours, needs reference dependence or narrow bracketing.

Required: MWG 6.A–6.C. **Next:** 6.D–6.F for Monday.

Applications: Rabin (2000); Sydnor (2010); Chetty (2006) optional.

Monday: ranking whole distributions, insurance, and what happens when probabilities themselves are unknown.